

Generic Singular-Reference Ref-GH Driver Implementation and Qualification Checkpoint

AthenaK Ref-GH development record

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Abstract

This report records the first implementation checkpoint for a generic puncture-adapted reference-frame generalized-harmonic system. A reference-independent first-order-state puncture-exponent estimator and a Gaussian-localized analytic singular reference were implemented and exercised locally. The analytic estimator and the prescribed reference-jet scan pass, but the required direct-finite-difference comparison fails for a scale invariance reason on the prescribed shell. Therefore closed-loop control is still **stopped at the initial estimator hard gate**. The subsequent standard γ_2 , hyperbolic gauge-driver, exact stationary-trumpet, and regular perturbed-trumpet local gates have advanced. No closed-loop control, wormhole evolution, SMR, or PVC qualification claim is made.

1 Provenance and scope

The mandated parent branch, its upstream-tracking reference, and the remote reference all equaled `0e248310a562c8a84327421eecf70f2f5d1da4a3` before editing. Work was performed on `codex/ref-gh-generic-singular-driver-20260825`; the Kokkos submodule is `6739bc623081648af9e752b616d9671527922cbf`. The local executable is a Release Kokkos Serial build with GCC 13.3.0. Existing work in other worktrees was not modified.

The three mechanisms remain conceptually separate:

Reference geometry	regularizes singular evolved components,
Hyperbolic gauge driver	controls the physical coordinate gauge,
γ_0/γ_2	controls GH and first-order constraints.

The reference/estimator slice, standard γ_2 additions, and physical hyperbolic gauge driver are implemented in this checkpoint.

2 Primary equations and convention map

Ordinary GH uses

$$C^a = \hat{H}^a + g^{bc}\Gamma^a_{bc}, \quad (1)$$

while the background-covariant form is

$$C^a = H^a_{\text{bg}} + g^{bc}(\Gamma^a_{bc} - \bar{\Gamma}^a_{bc}). \quad (2)$$

Equality of the physical constraint gives the sign relation

$$H^a_{\text{bg}} = \hat{H}^a + g^{bc}\bar{\Gamma}^a_{bc}. \quad (3)$$

Production reconstructs this relation from evolved ordinary $\hat{H}_A$ and the current reference geometry.

The runtime γ_2 additions and Einstein characteristic map implement Eqs. (35)–(37) and (32)–(34) of Ref. [1]. The conformal Gamma-driver with auxiliary Υ^i follows Eqs. (60)–(62) of Ref. [2]; the improved $\hat{H}_a, \theta_a$ driver follows Eqs. (9) and (11), with the combined characteristic system mapped from Appendix B of Ref. [3].

3 Pointwise physical exponent estimator

At each actual Cartesian cell center, with $X^i = x^i - x_{\text{puncture}}^i$, the implemented production quantity is

$$q_{\text{loc}} = -\frac{1}{6}X^k\gamma^{ij}\partial_k\gamma_{ij}. \quad (4)$$

The physical metric and derivative are reconstructed from Ψ , Φ , and the reference two-jets; no additional derivative is taken. The offline lapse diagnostic is

$$p_{\text{loc}} = X^k\partial_k \ln \alpha, \quad \alpha = (-g^{00})^{-1/2}. \quad (5)$$

The selection and weighting are

$$2h \leq r < 8h, \quad 8h < R_G/2, \quad w_i = (2h/r_i)^3, \quad (6)$$

with weighted variance and $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. No clipping, rays, radial interpolation, binning, or puncture extrapolation is used.

For wormhole data,

$$\psi_W = 1 + \frac{M}{2r}, \quad q_{\text{loc}}(r) = -2r \frac{d \ln \psi_W}{dr} = \frac{M}{r + M/2}. \quad (7)$$

The first-order-state calculation agrees pointwise to at most 1.6×10^{-15} . For the stationary trumpet it agrees with the independent profile derivative to at most 7.1×10^{-13} , and Minkowski is exactly zero.

The production estimate uses every valid shell point. For the independent FD comparison, a point is discarded if any fourth-order directional stencil spans a puncture coordinate, $|X^i| \leq 2h$. The state and FD values below use the same puncture-clear subset.

h/M	W prod.	W safe	W FD	T prod.	T safe	T FD
1/16	1.32184	1.11591	1.11851	0.856373	0.791536	0.792402
1/24	1.48531	1.30752	1.31097	0.902691	0.856383	0.857347
1/32	1.58484	1.43073	1.43477	0.926983	0.891313	0.892331
1/48	1.70032	1.58002	1.58484	0.951826	0.927678	0.928754
1/64	1.76543	1.66723	1.67252	0.964342	0.946240	0.947345

Table 1: Weighted estimates on the required shell. The full production sample has 2,144 points and $N_{\text{eff}} = 595.547$; the safe comparison has 480 points and $N_{\text{eff}} = 370.885$.

4 Direct-FD hard-gate failure

The independent diagnostic reconstructs γ_{ij} on neighboring actual cell centers and applies AthenaK’s fourth-order centered operator directly. The puncture exclusion substantially reduces

the state–FD discrepancy, but it still grows from 0.00260102 to 0.00528934 for the wormhole and from 0.000865660 to 0.00110559 for the trumpet. The strict executable therefore exits nonzero.

This behavior follows from scale invariance. For $\gamma_{ij} = r^{-2q}\delta_{ij}$, write a selected cell as $x^i = h\xi^i$. Any fixed centered stencil obeys

$$D_h\gamma(h\xi) = h^{-2q-1}D_1\gamma(\xi). \quad (8)$$

Because $X^k = h\xi^k$ and $\gamma^{ij} = h^{2q}\delta^{ij}$, the discrete contraction $X^k\gamma^{ij}D_k\gamma_{ij}$ is independent of h at fixed r/h . Unless the stencil is exact for the singular power, the bias cannot converge away. The weighted pure-power limits for this stencil and shell are about 0.989337 for $q = 1$ and 1.575776 for $q = 2$.

Thus the claim “local puncture-exponent estimator established” is not made under the controlling same-shell comparison, despite the analytic success of the first-order-state path.

5 Standard γ_0/γ_2 checkpoint

Runtime `gamma2` defaults to zero. For fixed $\gamma_1 = -1$, production now adds

$$\delta\Pi_{AB,t} = -\gamma_2\beta^I C_{IAB}, \quad \delta\Phi_{IAB,t} = \alpha\gamma_2 C_{IAB}. \quad (9)$$

The device gate checks the forward/inverse characteristic map $u^{1\pm} = \Pi \pm s^I\Phi_I - \gamma_2\Psi$, the standard symmetrizer, and the frozen reduction and curl subsidiary damping to 3.33×10^{-16} .

A six-case periodic runtime matrix through $t = 0.2$ covers $\gamma_2 = 0, 0.5, 1$ with KO off and on. Reduction and curl histories obey

$$\|C(t)\|_2 = \|C_{\gamma_2=0}(t)\|_2 e^{-\gamma_2 t} \quad (10)$$

with maximum absolute growth-factor errors 7.07×10^{-7} and 2.87×10^{-6} . The $\gamma_2 = 0$ histories independently measure the KO contribution.

For γ_0 , a reduction-clean transverse GH-constraint eigenmode is built from Eq. (21) of Ref. [1]. Matrices at $\gamma_0 = 0.25, 0.5, 1$, with KO off and on, obey

$$\|C_y(t)\|_2 = \|C_y(0)\|_2 e^{-(\gamma_0/2 + \lambda_{\text{KO}})t} \quad (11)$$

to 5.35×10^{-7} . The largest accompanying reduction norm is 7.19×10^{-18} and curl is zero. These are local linearized/robust subsidiary tests, not nonlinear black-hole, long-time, SMR, or GPU qualification.

6 Hyperbolic gauge and static-reference regularization

The evolved state now includes four $\widehat{H}_A$, four θ_A , and three Υ^i fields. Exact frame-motion terms, the advective 1+log target, the conformal Gamma-driver target, nonzero- $\widehat{H}$ source terms, combined characteristics, communication, output, and restart paths are included.

Raw ordinary-gauge frame components are singular at the puncture. For static references the stored variables are therefore

$$\delta\widehat{H}_A = \widehat{H}_A - \widehat{H}_A^{\text{ref}}, \quad \delta\theta_A = \theta_A - \theta_A^{\text{ref}}. \quad (12)$$

Physical sources reconstruct the unchanged raw fields. With this equation-preserving representation, the exact trumpet is a roundoff fixed point through $t = 1$ at 16^3 , 24^3 , and 32^3 . Time-dependent-reference subtraction still fails closed pending an analytic time derivative of the θ baseline.

For fourth-order Ref-GH the centered derivative radius is two cells, while the enabled KO operator reaches three. All evolved-state histories and convergence results now discard the full radius-three support. The offline tensor-product comparison rejects a target if any contributing source cell has a support box containing the puncture. A current-build replay changes only the included diagnostic volume; its field and constraint arrays are elementwise identical to the pre-correction payload.

For a regular $r^8 \exp(-r^2/w^2)$ perturbation, the masked $24^3/32^3/48^3$ self-orders at $t = 1$ are 4.700 for field L2, 4.742 for field Linf, 3.948 for native-constraint L2, and 3.015 for native-constraint Linf. All three cases remain finite. Exact-time L2 orders between $t = 0.4$ and $t = 1$ are resolution improving and approximately, but not uniformly, fourth order. The outer faces at $2M$ cannot reach the $r < 1M$ analysis region before $t \simeq 1.64M$ at the measured maximum characteristic speeds.

7 Gaussian singular reference and prescribed trajectory

The new positive isotropic Cholesky factor is

$$\sigma_q(r, t) = \exp[-q(t)W(r) \ln(r/M)], \quad W(r) = \exp[-(r/R_G)^2]. \quad (13)$$

A quintic endpoint-flat $q : 2 \rightarrow 1$ trajectory supplies analytic value, first derivative, and second derivative. Consequently the reference jets retain both

$$\partial_t \ln \sigma_q = -\dot{q}W \ln(r/M) \quad (14)$$

and the $\ddot{q}W \ln(r/M)$ and $\dot{q}^2 W^2 \ln^2(r/M)$ second-time-derivative structures.

The scan covers $\tau/M = 4, 8, 16$, $R_G/M = 2, 3, 4$, and $h/M = 1/16, 1/24, 1/32, 1/48, 1/64, 1/128$. Each dynamic midpoint has a static $q = 1.5$ control. The staged production cache agrees with the full geometry oracle to 1.11×10^{-15} at $t = \tau/2$.

Measure, $\tau = 8M, R_G = 3M$	$h = M/16$	$h = M/128$	model
$ \dot{q}W \ln \rho $	0.683316	1.170901	log
$\dot{q}^2 W^2 \ln^2 \rho$	0.466920	1.371010	log-squared
reference Ricci maximum	1.42809	4.11641	log-squared
total source maximum	2.85617	8.23283	log-squared

Table 2: Representative prescribed reference-jet scaling.

The compact analyzer finds no dynamic measure classified as unexpected positive algebraic growth. The prescribed- $q(t)$ reference-jet gate therefore passes locally. This result says nothing about physical evolution or feedback.

8 Qualification matrix and stop condition

Gate	Status
First-order-state estimator, analytic initial geometries	Passed locally
Direct-FD same-shell convergence	Failed
Prescribed Gaussian two-jets and staged cache	Passed locally on Kokkos Serial
Runtime γ_0/γ_2 GH, reduction, curl subsidiary matrices	Passed locally
Closed-loop q control	Not started; blocked by estimator gate
Dynamic $\hat{H}/\theta/\Upsilon$ gauge system	Implemented; local oracles pass
Exact stationary trumpet through $t = 1$	Roundoff fixed point locally
Regular perturbed trumpet through $t = 1$	Finite; approximately fourth-order masked L2
Focused 61-field restart	Passed locally
Generic/time-dependent gauge, wormhole, SMR, CUDA, Aurora PVC	Not established

No Aurora job was submitted in this checkpoint: the SSH ControlMaster socket is absent and noninteractive authentication fails. Local B580 Level Zero enumeration also hung, so no device claim is made. Before feedback work, the specification must choose between a fixed-physical-annulus direct-FD convergence check or a fixed- r/h known-bias comparison. Both avoid interpolation; neither is the literal present demand.

References

- [1] L. Lindblom, M. A. Scheel, L. E. Kidder, R. Owen, and O. Rinne, “A New Generalized Harmonic Evolution System,” [arXiv:gr-qc/0512093](#).
- [2] L. Lindblom, K. D. Matthews, O. Rinne, and M. A. Scheel, “Gauge Drivers for the Generalized Harmonic Einstein Equations,” [arXiv:0711.2084](#).
- [3] L. Lindblom and B. Szilagyi, “An Improved Gauge Driver for the Generalized Harmonic Einstein System,” [arXiv:0904.4873](#).